

Exercises 1.3

1. $H \cap J$ is a subgroup of H and of J . Thus $|H \cap J|$ divides both $|H|$ and $|J|$, so it can be 1, 2 or 4.

$$|HJ| = \frac{12 \times 8}{|H \cap J|}. \text{ This could be 24, 48 or 96.}$$

HJ is not a group unless HJ and JH happen to be equal.

2. The given conditions imply $|H| \cdot |J| > \sqrt{|G|} \sqrt{|G|} = |G|$. As $HJ \subset G$, this gives $|H| \cdot |J| > |HJ|$. Hence by Proposition 1.4, $|H \cap J| > 1$.

Suppose a group of order 15 has subgroups H and J of order 5. As $5^2 > 15$, we have $|H \cap J| > 1$ by the above. As $|H \cap J|$ divides 5 and is not 1, it must equal 5, i.e. $H \cap J = H = J$. Thus there is at most one 5-element subgroup.

3. As G is not cyclic, no element has order 55. Only the identity has order 1 so the remaining elements all have order 5 or 11, hence they all generate subgroups of those prime orders.

Each of the r 5-element subgroups contains 4 elements of order 5 and the identity. Each of the s 11-element subgroups contains 10 elements of order 11 and the identity. Thus $4r + 10s + 1 = 55$, so $4r + 10s = 54$, so $2r + 5s = 27$.

Hence (r, s) can be $(1, 5)$, $(6, 3)$ or $(11, 1)$.

The numbers of elements of order 5 and 11 are then respectively

$(4, 50)$, $(24, 30)$, $(44, 10)$.

(We shall see in Chapter 2 that only one of these cases can actually occur.)

4. (a) If $N \triangleleft G$ then $aba^{-1} \in N$ for all $a \in G, b \in N$.

If H is a subgroup of G which contains N then N is clearly a subgroup of H . If $h \in H$ then $h \in G$ so by the above, $hnh^{-1} \in N$ for all $h \in H, n \in N$. Thus $N \triangleleft H$.

- (b) In D_4 take $N = \{\iota, \sigma^2, \tau, \sigma^2\tau\}$ and $H = \{\iota, \tau\}$. Then $N \triangleleft D_4$ and $H \triangleleft N$, but as shown in the notes H is not normal in G .

5. C_6/H consists of the three cosets $H = \{e, g^3\}$, $gH = \{g, g^4\}$, $g^2H = \{g^2, g^5\}$, so the

quotient group is isomorphic to C_3 with table

	H	gH	g^2H
H	H	gH	g^2H
gH	gH	g^2H	H
g^2H	g^2H	H	gH

C_8/H consists of the two cosets $H = \{e, g^2, g^4, g^6\}$ and $gH = \{g, g^3, g^5, g^7\}$, so the

quotient group is isomorphic to C_2 with table

	H	gH
H	H	gH
gH	gH	H

6. (a) S_3/A consists of the two cosets $A = \{\iota, \sigma, \sigma^2\}$ and $\tau A = \{\tau, \sigma\tau, \sigma^2\tau\}$, so the

quotient group is isomorphic to C_2 with table

	A	τA
A	A	τA
τA	τA	A

- (b) $AB = \{\iota, \sigma, \sigma^2\}\{\iota, \tau\} = \{\iota, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} = S_3$.

However, $A \times B \cong C_3 \times C_2 \cong C_6$ which is abelian but S_3 is non-abelian, so $S_3 \not\cong A \times B$. B is not normal in S_3 (and $\sigma\tau \neq \tau\sigma$).

7. (a) Suppose G is abelian and let $aN, bN \in G/N$. Then $(aN)(bN) = abN = baN = (bN)(aN)$, so G/N is abelian.
- (b) Suppose $G = \langle g \rangle = \{g^r : r \in \mathbb{Z}\}$. Then $G/N = \{aN : a \in G\} = \{g^r N : r \in \mathbb{Z}\} = \{(gN)^r : r \in \mathbb{Z}\} = \langle gN \rangle$, so G/N is cyclic, generated by gN .
- (c) If aN has order m in G/N then $(aN)^m$ is the identity in G/N , which is N . Thus $a^m N = N$. This says that the coset of N that contains a^m is N itself, i.e. $a^m \in N$.

8. We see that $X^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. As $5 = 0$ in $\mathbb{F}_5$, we have $X^5 = I_2$ so X has order 5.

Also $Y^n = \begin{pmatrix} 2^n & 0 \\ 0 & 1 \end{pmatrix}$. As $2^4 = 1$, i.e. 2 has order 4 in $\mathbb{Z}_5^*$, we have $Y^4 = I_2$ so Y has order 4.

$2^{-1} = 3$ in $\mathbb{F}_5$ (as $2 \times_5 3 = 1$) so $YXY^{-1} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = X^2$. Thus $YX = X^2Y$, so the elements of $\langle X, Y \rangle$ are all products $X^m Y^n$ where $m \in \{0, \dots, 4\}$, $n \in \{0, \dots, 3\}$, i.e. $\langle X, Y \rangle = \langle X \rangle \langle Y \rangle$.

Clearly $\langle X \rangle \cap \langle Y \rangle = \{I_2\}$, so $\langle X, Y : X^5 = Y^4 = I, YXY^{-1} = X^2 \rangle$ is a group of order $5 \times 4 = 20$.

Taking X as given and $Y = \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$ where $g \in \{2, \dots, p-2\}$ generates $\mathbb{Z}_p^*$, we get a group of order $p(p-1)$ for each prime p , with presentation $\langle X, Y : X^p = Y^{p-1} = I, YXY^{-1} = X^g \rangle$.

9. H has index 2 in D_6 , so $H \triangleleft D_6$.

$D_6 = \langle a, b \rangle$ and $J = \langle a^3 \rangle$. Clearly $aa^3a^{-1} = a^3 \in J$.

From the presentation we get $bab^{-1} = a^{-1}$.

Thus $ba^3b^{-1} = (bab^{-1})^3 = (a^{-1})^3 = (a^3)^{-1} \in J$.

Hence $J \triangleleft D_6$ by Proposition 1.1.

Letting $c = a^2$ we have $c^3 = b^2 = e$ and $cbc = a(aba)a = aba = b$ so H satisfies the presentation of D_3 , i.e. $H \cong D_3$. Clearly $J \cong C_2$.

Also $H \cap J = \{e\}$ so $|HJ| = |H| \cdot |J| = 12$. As $|D_6| = 12$ and $HJ \subset D_6$, we have $HJ = D_6$. It follows by Proposition 1.5 (ii) that $D_6 \cong H \times J \cong D_3 \times C_2$.

10. Let $H = \{e, \sigma^2\} \triangleleft D_4$. Then H and D_4/H are abelian, so $H \times D_4/H$ is abelian, but D_4 is non-abelian so $D_4 \not\cong H \times D_4/H$.

Let $H = \{e, g^3\} \triangleleft C_6$. Then $H \cong C_2$, $C_6/H \cong C_3$ and $C_6 \cong C_2 \times C_3$, so $C_6 \cong H \times C_6/H$.

11. (a) Conjugating the generator a^2 of H by each of the generators a, b of Q_4 gives the same set H , so $H \triangleleft G$.

The cosets are $H = \{e, a^2\}$, $aH = \{a, a^3\}$, $bH = \{b, a^2b\}$, $abH = \{ab, a^3b\}$.

These form the quotient group Q_4/H which we find to be isomorphic to V , but Q_4 has only two self-inverse elements so it cannot have a subgroup isomorphic to V .

(b) Using Proposition 1.7, the non-trivial proper subgroups of Q_4/H are $\{H, aH\}$, $\{H, bH\}$ and $\{H, abH\}$.

12. If $g_1N = g_2N$ then $g_1^{-1}g_2 \in N$, so $g_1^{-1}g_2 \in H$, so $g_1H = g_2H$. Hence ψ is well-defined, i.e. independent of how a coset is represented.

$\psi(g_1Ng_2N) = \psi(g_1g_2N) = g_1g_2H = (g_1H)(g_2H) = \psi(g_1N)\psi(g_2N)$, so ψ is a group homomorphism.

$\psi(G/N) = \{gH : g \in G\} = G/H$, i.e. ψ is surjective.

$\text{Ker } \psi = \{gN \in G/N : gH = H\} = \{gN \in G/N : g \in H\} = H/N$.

Thus $H/N \triangleleft G/N$. By the first isomorphism theorem, $(G/N)/(H/N) \cong G/H$.

13. $D_4/H \cong V$ and D_4 has subgroups isomorphic to V , e.g. $\{\iota, \sigma^2, \tau, \sigma^2\tau\}$ (see table on Page 5 of notes). D_4 is not cyclic, so it is *not* the case that if G has a subgroup isomorphic to G/N then G/N is cyclic.

14. (a) By the first isomorphism theorem, $G_1/\text{Ker } \phi \cong \phi(G_1)$, so $|G_1| = |\text{Ker } \phi| \cdot |\phi(G_1)|$. These orders are clearly integers, so $|\phi(G_1)|$ divides $|G_1|$.

Also $\phi(G_1)$ is a subgroup of G_2 , so by Lagrange's theorem $|\phi(G_1)|$ divides $|G_2|$.

- (b) $|S_3| = 6$ and $|C_7| = 7$, so if $\phi : S_3 \rightarrow C_7$ is a group homomorphism then $|\phi(S_3)|$ divides both 6 and 7. Thus $|\phi(S_3)| = 1$, i.e. $\phi(S_3) = \{e\}$, so ϕ is the trivial map.

- (c) $\phi(\langle a \rangle) = \phi(a^k : k \in \mathbb{Z}) = \{\phi(a)^k : a \in \mathbb{Z}\} = \langle \phi(a) \rangle$. Applying part (a) to the map $\phi : \langle a \rangle \rightarrow \langle \phi(a) \rangle$ shows that $|\langle \phi(a) \rangle|$ divides $|\langle a \rangle|$, i.e. the order of $\phi(a)$ divides the order of a .

If ϕ is injective then $|\langle \phi(a) \rangle| = |\langle a \rangle|$, so $\phi(a)$ and a have the same order.

15. N is a subgroup of G . It is contained in HN (since $n = en$ for all $n \in N$, and $e \in H$), so N a subgroup of HN . It is closed under conjugation by all elements of G , thus in particular by those in HN , so $N \triangleleft HN$.

Define $\phi : H \rightarrow HN/N$ by $\phi(h) = hN$. (Note that the codomain is HN/N because H/N is not meaningful unless $N \subset H$.)

For any $h_1, h_2 \in H$, $\phi(h_1h_2) = h_1h_2N = (h_1N)(h_2N)$, so ϕ is a group homomorphism.

$\text{Ker } \phi = \{h \in H : \phi(h) = e_{HN/N}\} = \{h \in H : hN = N\} = \{h \in H : h \in N\} = H \cap N$.

$\phi(H) = \{hN : h \in H\} = \{hnN : h \in H, n \in N\} = HN/N$, so ϕ is surjective.

By the first isomorphism theorem, $H/(H \cap N) \cong HN/N$.